

1. 通讯协议与故障报文 Communication protocol and fault message

波特率: 500kbps(可选250kbps) Baud rate: 500kbps (Optional 250kbps)
 采样点: 75% Sampling Point: 75%
 帧格式: 扩展帧 Frame Format: Extended Frame
 数据格式: Motorola格式 Data Format: Motorola Format

1: BMS to TMS

OUT	IN	ID (0x18FF45F4)						周期 Cycle(ms)
BMS	所有节点 All Nodes	P	R	DP	PF	PS	SA	1000
		6	0	0	FF	45(69)	F4(244)	
位置(字节/位) Location (Byte/Bit)	参数定义 Parameter Definition	参数说明 Parameter Description						
1.2-1.1	BMS 请求模式 BMS Request Mode	0: 关机模式 Shutdown Mode 1: 制冷模式; Cooling Mode 2: 制热模式; Heating Mode 3: 自循环模式 Self Recycling Mode						
1.4-1.3	高压请求 High Voltage Request	0: 上高压请求 HV On 1: 下高压请求 HV Off 2~3: 无效 Invalid						
1.6-1.5	充电状态 Charging Mode	0: 非充电模式 Non-charging mode 1: 充电模式 Charging mode 2~3: 无效 Invalid						
1.8-1.7	BMS 高压继电器状态 BMS High Voltage Relay Status	0: 断开状态 Open 1: 闭合状态 Close 2~3: 无效 Invalid						
2	BMS 当前电压高字节 BMS Current Voltage High Byte	分辨率: 1V / bit Resolution ratio: 1V / bit 偏移量: 0 Offset: 0 范围: 0V~1000V Range: 0V~1000V						
3	BMS 当前电压低字节 BMS Current Voltage Low Byte	类型: 测量 Type: Measurement 65535: 无效 65535: Invalid						
4	保留 Reserve	保留 Reserve						
5	设定温度 Setting Temperature	分辨率: 1° C/bit Resolution ratio: 1° C/bit 偏移量: -40° C Offset: -40° C 单位: ° C Offset: -40° C						
6	保留 Reserve	保留 Reserve						
7	BMS 生命帧 Life Frame	0~255						
8	CRC 校验值 Check Value	0~255						

注: 添加CRC 校验值作为备选项

Note: Add CRC check value as options;

2: TMS to BMS

OUT	IN	ID (0x18FFC13A)						周期 Cycle (ms)
TMS	所有节点 All Nodes	P	R	DP	PF	PS	SA	1000
		6	0	0	FF	C1 (193)	3A (58)	
位置 (字节/位) Location (Byte /Bit)	参数定义 Parameter Definition	参数说明 Parameter Description						
1. 2-1. 1	TMS 工作状态 TMS Working Status	0: 关机模式 Shutdown Mode 1: 制冷模式; Cooling Mode 2: 制热模式; Heating Mode 3: 自循环模式 Self Recycling Mode						
1. 4-1. 3	TMS 高压继电器状态 TMS High Voltage Relay Status	0: 断开状态 Open 1: 闭合状态 Close 2~3: 无效 Invalid						
1. 8-1. 5	保留 Reserve	保留 Reserve						
2	出水温度 (机组到电池) Output Water Temperature (TMS to Battery)	分辨率: 1° C/bit Resolution Ratio: 1°C/bit 偏移量: -40° C Offset: -40°C 范围: -40°C~210°C Range: -40°C~210°C 单位: ° C Unit: °C 类型: 测量 Type: Measurement 255: 无效 255: Invalid						
3	回水温度 (电池到机组) Return Water temperature (Battery to TMS)	分辨率: 1° C/bit Resolution Ratio: 1°C/bit 偏移量: -40° C Offset: -40°C 范围: -40°C~210°C Range: -40°C~210°C 单位: ° C Unit: °C 类型: 测量 Type: Measurement 255: 无效 255: Invalid						
4	室外环境温度 Environment temperature(TMS to Battery)	分辨率: 1° C/bit Resolution Ratio: 1°C/bit 偏移量: -40° C Offset: -40°C 范围: -40°C~210°C Range: -40°C~210°C 单位: ° C Unit: °C 类型: 测量 Type: Measurement 255: 无效 255: Invalid						
6	TMS需求功率高字节 TMS Demand Power High Byte	分辨率0.1Kw/bit Resolution: 0.1 Kw/bit 偏移量: 0 Offset: 0 范围0Kw~100.0Kw Range: 0Kw~100.0Kw 类型: 测量 Type: Measurement 65535: 无效 65535: Invalid						
7	TMS需求功率低字节 TMS Demand Power High Byte							
8. 6-8. 1	TMS故障码 TMS Error Code	0: 无故障 有关其他特定故障代码, 请参见故障代码表 0: No error For other specific error codes, please check the error code list						
8. 8-8. 7	TMS故障等级 TMS Error Level	0: 无故障 0: No Error 1: 1 级故障 1: Level 1 Error 2: 2 级故障 2: Level 2 Error						

注：Note:

1) TMS故障码出现多个时采用循环发送方式。

1) When there are multiple TMS error codes, the cyclic transmission method is adopted.

2) 故障等级分为2级：自定义1级、2级故障，其中1级等级最高。多个故障发生时，只发送最高等级。

2) The error level is divided into 2 levels: custom level 1 and level 2 errors, of which level 1 is the highest. When multiple errors occur, only the highest level is sent.

例如：TMS中同时存在1级和2级故障时，TMS应保持发送1级故障等级，并循环发送故障码。

For example: when there are both level 1 and level 2 errors in the TMS, the TMS should keep sending the level 1 error and send the error code cyclically.

3) 故障处理：BMS只响应故障等级，故障处理方式如下：

3) Troubleshooting: BMS only responds to the error level, and the error handling method is as follows:

1级故障处理方式：发送关机指令、请求下高压指令、断高压、转发故障给整车；

Level 1 Troubleshooting: Send shutdown commands, request high-voltage off commands, cut off high-voltage, and forward errors to the vehicle;

2级故障处理方式：发送关机指令、转发故障给整车；

Level 1 Troubleshooting: Send shutdown commands, request high-voltage off commands, cut off high-voltage, and forward errors to the vehicle;

故障代码表 error code list

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
1		无故障 Trouble-free		
2	48	BMS 和TMS “CAN” 通讯异常 BMS and TMS "CAN" communication abnormality	机组不受控制（判断时间需 $\leq 7s$ ） The unit is out of control (judgment time needs to be $\leq 7s$)	Level1
3	1	PTC 1 级故障 PTC level 1 fault	IGBT故障/或IGBT 击穿 IGBT failure/or IGBT breakdown	Level1
4	2	PTC 2级故障 PTC level 2 fault	PTC 过压 PTC overvoltage	Level2
5	5	PTC 2级故障 PTC level 2 fault	PTC 欠压 PTC undervoltage	Level2
6	11	PTC 2级故障 PTC level 2 fault	NTC坏/或PTC自身故障 NTC damaged/or PTC itself faulty	Level2
7	14	PTC 2级故障 PTC level 2 fault	PTC通讯故障（或生命帧停止更新时间大于10秒） PTC communication failure (or life frame stops updating for more than 10 seconds)	Level2

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
8	19	AC 2 级故障 AC level 2 fault	压缩机故障（制冷模式） Compressor failure (cooling mode)	Level2
9	20	BMS高压供电故障 BMS high voltage power supply failure	在制冷或制热工作时需要使用高压设备(压缩机或PTC)时,BMS提供的电压过高或过低。 When high-voltage equipment (compressor or PTC) is required for cooling or heating work, the voltage provided by the BMS is too high or too low.	Level2
10	21	Fan 2 级故障 Fan level 2 fault	冷凝风扇故障电源故障 Condenser fan failure Power supply failure	Level2
11	22	高压传感器2 级故障 High voltage sensor level 2 fault	压缩机高压过高，过低或者高压传感器开路，短路，系统压力过高（制冷模式） The compressor high pressure is too high or too low or the high pressure sensor is open or short circuited, and the system pressure is too high (cooling mode)	Level2
12	23	低压传感器2 级故障 Low voltage sensor level 2 fault	压缩机低压过高，过低或者低压传感器开路，短路，系统压力过低（制冷模式） The compressor low pressure is too high or too low or the low pressure sensor is open or short circuited, and the system pressure is too low (cooling mode)	Level2
13	24	机组进水温度传感器 2 级故障 Unit inlet water temperature sensor Level 2 fault	机组进水温度传感器故障，开路，短路（全模式） The unit inlet water temperature sensor is faulty, open circuit, short circuit (all modes)	Level2
14	25	机组出水温度传感器 2 级故障 Unit outlet water temperature sensor Level 2 fault	机组出水温度传感器故障，开路，短路（全模式） The unit outlet water temperature sensor is faulty, open circuit, short circuit (all modes)	Level2
15	26	机组环境温度传感器 2 级故障 Unit ambient temperature sensor Level 2 fault	机组环境温度传感器故障，开路，短路（全模式） Unit ambient temperature sensor failure, open circuit, short circuit (all modes)	Level2
16	27	水泵 water pump	水泵电源故障 Water pump power failure	Level2
17	30	水位传感器2 级故障 Water level sensor	水位传感器故障，开路或者短路（全模式） Water level sensor failure, open circuit or short	Level2

序号 S/N	故障代码 Error Code	故障内容 Fault Content	造成原因 Cause	故障等级 Fault Level
		level 2 fault	circuit (all modes)	
18	31	PTC 2级故障 PTC Level 2 fault	PTC出水温度高/或PTC过温故障 PTC outlet water temperature is high/or PTC over temperature fault	Level2
19	47	低压供电 2 级故障 Low voltage power supply level 2 fault	低压供电低于 18V 或者高于 32V Low voltage power supply is lower than 18V or higher than 32V	Level2
20	51	机组防冻温度传感器 2 级故障 Unit anti-freeze temperature sensor level 2 fault	机组防冻温度传感器故障, 开路, 短路 (全模式) Unit antifreeze temperature sensor failure, open circuit, short circuit (all modes)	Level2

注：如果设备为冷暖协议单冷型，则可以忽略制热部分功能及故障反馈。

Note: If the device is a single cooling type with a heating and cooling protocol, the heating function and fault feedback can be ignored.